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5. Fig. 6(a) 的曲线趋势与正文物理解释不一致

正文写道：

The model achieves optimal performance near the RIS normal direction. However, as the angle deviates from the normal, the MSE gradually increases...

但 Fig. 6(a) 实际曲线表现为：

- 从约 0° 到 100° ，MSE 从约 5.4×10^{-5} 下降到约 0.5×10^{-5} ；
- 从约 260° 到 350° ，MSE 又从约 0.5×10^{-5} 上升到约 5.6×10^{-5} 。

如果你定义的 RIS normal direction 是 $0^\circ/360^\circ$ ，图中反而是在法向附近误差最大，与正文完全相反。

这里必须先检查生成该图的角度定义：

- 如果 0° 是 RIS 法向，图的数据或正文解释至少有一个错误；
- 如果 90° 或其他角度是法向，必须在正文或图注中明确说明角度参考方向。

在没有确认角度定义前，不建议保留“near the RIS normal direction”这一物理结论。可以暂时改成纯数据描述：

The channel-estimation MSE varies significantly with the target angle and remains relatively low near the boundaries of the observable angular regions.

不过最稳妥的是先检查绘图代码中的角度坐标定义。

1. 相同默认配置下的 sensing EE 出现了多组不同结果

第 18 页默认参数为：

- $M = 8$
- $N = 64$
- $K = 2$
- $K_R = 5$
- $P_t = 30$ dBm

但不同图表中，默认或接近默认条件下的 proposed-method EE 大约为：

位置	条件	图中结果
Fig. 5	训练收敛后的 EE	约 85 dB
Fig. 7	$P_t = 30$ dBm, Proposed Two-Stage	约 83.2 dB
Fig. 10	$P_t = 30$ dBm, Deep Learning	约 77.5 dB
Fig. 13(a)	$K_R = 5$ nominal	约 81.8 dB

五、与公式和实验结果直接相关的其他重要一致性问题

1. 综合损失函数没有 sensing-SNR penalty, 但后文报告 sensing-SNR satisfaction rate

当前公式 (19):

$$L_{\text{total}} = -EE_s + \lambda_1 L_{\text{smooth}} + \lambda_2 L_{\text{power}} + \lambda_3 L_{\text{SNR}}.$$

其中没有 L_{SNR} , 但:

- 问题 (16e) 明确要求 $\text{SNR}_r \geq \tau_r$;
- 第 31 页又报告 sensing-SNR satisfaction rate = 99.1%;
- 正文声称在保证 sensing requirement 的情况下训练。

这会使审稿人追问: 模型如何在训练中显式处理 (16e)?

建议增加:

$$+ \lambda_4 L_{\text{SNR}},$$

例如:

$$L_{\text{SNR}} = \frac{1}{B} \sum_{i=1}^B \text{ReLU}(\tau_r - \text{SNR}_{r,i}).$$

随后权重敏感性分析也要说明 λ_4 的设置。若不准增加该项, 就不能声称 loss 显式保证 sensing-SN constraint, 只能说实验中统计到较高满足率。

当前公式 (16e) 和公式 (19) 中的 N 不一致。

$$2(NM + N),$$

不是 $2(NM + M)$ 。

当前 $M = 8, N = 64$ 时:

$$2(NM + N) = 2(64 \times 8 + 64) = 1152,$$

所以图中的 1152 实际对应 $N = 64$ 。Fig. 2 的当前数值和公式标签确实体现了 $N = 64$ 的设置。 [DLS](#)

如果改成 $N = 32$, 则:

$$2(NM + N) = 2(32 \times 8 + 32) = 576.$$

因此 Fig. 2 中需要同时修改:

当前

$$2(NM + M)$$

修改后

Fig. 13(b): 仍然没有改成 $|\epsilon|$

当前正文仍写:

slightly decreases as the spacing perturbation increases.

但从 -10% 到 0 , EE是上升的; 从 0 到 $+10\%$ 才下降。

应改成:

The sensing EE reaches its maximum at the nominal spacing $d = \lambda/2$ and gradually decreases as the absolute spacing perturbation $|\epsilon|$ increases.

完整建议:

Fig. 13(b) shows that the logarithmic sensing EE reaches its maximum at the nominal array spacing $d = \lambda/2$, corresponding to $\epsilon = 0$. When the array spacing is either compressed or expanded, the sensing EE gradually decreases as the absolute perturbation magnitude $|\epsilon|$ increases.

2. EE 实验结果存在明显冲突, 必须统一重跑或重新导出

目前至少存在以下几组冲突。

2.1 Fig. 8 图中数值与正文不一致

Fig. 8(c) 图内标注为:

MAX EE range: 79.32–84.54 dB/J

但正文写的是:

88.53 dB 降至 83.29 dB。

而且图中仍然保留了审稿人要求删除的 **dB/J**。  DL8 +1

2.4 不同图中的默认 EE 水平不统一

目前大致存在三组默认结果：

- Fig. 3、Table II、Fig. 13：约 81.9 dB；
- Fig. 7、Fig. 10：约 83–84 dB；
- Fig. 5、Fig. 8、Fig. 12：约 85–88 dB。

如果这些实验使用相同的默认参数 $N = 32$ 、 $P_{\max} = 30$ dBm、 $K = 2$ 、 $\tau_r = 15$ dB，就会有如此大的差异。

2. 引用 [4] 与正文内容不对应

引言用 [4] 支撑“NLoS 导致目标检测和通信性能下降”，但 [4] 是关于 sliced C-RAN、UeMBB revenue maximization 的论文，与 NLoS/ISAC target detection 没有直接关系。必须换成真正研究 NLoS 对 ISAC sensing/communication 性能影响的文献。

3. 回复信参考文献编号没有随正文更新

典型冲突包括：

- 回复信 R1-C1：[18]–[21]；正文对应 [20]、[21]、[25]、[26]；
- 回复信 R1-C2：[22]；正文为 [27]；
- 回复信 R1-C3：SLBM [33]、PF [34]；正文为 [37]、[38]；
- 回复信 R1-C6：BCD [11],[29]、SLBM [33]、PF [34]；正文为 [11],[34],[37],[38]；
- 回复信 R2-C2：According to [28]；正文式 (12)–(13) 为 [33]；
- 回复信 R2-C7：[24]–[26]；正文实际是 [22]–[24]。 

最终参考文献排序固定以后，需要对回复信中的所有编号进行一次全局替换。

Abstract/Introduction

- cascade channel → cascaded channel
- the requirement of the data communication → the demand for data communication
- for the 6G → for 6G
- some key parameters optimizations → key optimization tasks
- sensing communication tradeoff → sensing-communication tradeoff
- shows the stronger generalization ability 语法和证据都不足

System model

- a target user (Target) → a passive sensing target
- signals which contains → a unified signal that contains
- there is no direct connection ..., a RIS is needed 是逗号连接句，应拆分
- The BS using M antennas → The BS equipped with M antennas
- user equipments → user equipment devices 或直接 K single-antenna UEs

Network section

- which are used the received signal → which use the received signal
- The third one is solving the periodic problem 表意错误
- an two-stage framework → a two-stage framework
- physics-inspired residual design 缺少具体物理机制，不宜直接使用
- avoids explicit estimation of the CSI 与全文“position-aware CSI construction”冲突

Simulation section

- To illustrate ..., To evaluate ... 重复
- [38]determines 缺少空格
- peformance → performance
- estimation → estimation
- lower user SINR Second 缺少句号
- numbers of RIS elements → number of RIS elements
- The performance of these four methods show → shows
- 3.4-3.2 bit/s/Hz, The proposed 标点错误

Conclusion

- an unknown position target → a target with an unknown position
- through evaluating the position → by estimating the target position
- designs beamforming matrix and optimize phase shift → designs the beamforming optimizes the RIS phase shifts
- end-to-end mapping from CSI 与两阶段未知目标信息流冲突
- “8–12 dB”与正文 Fig. 12 的“8.5–16 dB”不一致；
- “three-order-of-magnitude”与正文“3–4 orders”不一致。  DL8

给 Reviewer 3 Comment 7 的完整回复

建议将你现在的回复整体替换成下面这一版：

Response: Thank you for carefully pointing out these issues. Regarding the power constraint, Eq. (2) presents the general transmission model including both communication signals and a dedicated radar signal. However, the sensing-EE optimization considered in this work is performed in the ISAC dual-function phase, where no dedicated radar waveform is transmitted. Therefore, $\mathbf{s}_0 = \mathbf{0}$, which gives $\mathbf{W}_0 = \mathbb{E}\{\mathbf{s}_0\mathbf{s}_0^H\} = \mathbf{0}$ and $\text{Tr}(\mathbf{W}_0) = 0$. We have clarified this relationship after Eq. (2), and the power constraint in (16c) is written in its reduced form as

$$\sum_{k=1}^K \|\mathbf{w}_k\|_2^2 \leq P_{\max}.$$

We have also replaced the previous sentence containing the undefined symbols d and θ_r . The revised text now states that the target-related geometric parameters, such as the RIS–target distance and angle, are unavailable because the target location is unknown, and consequently the RIS–target channel $\mathbf{h}$ and the round-trip sensing channel $\mathbf{B}(\boldsymbol{\theta})$ cannot be directly obtained. In addition, the typographical errors “assmued,” “appling,” and “By appliing similar optimization logic” have been corrected throughout the manuscript.

然后粘贴正文修改：

In the ISAC dual-function phase considered in this work, no dedicated radar sensing signal is transmitted. Therefore, $\mathbf{s}_0 = \mathbf{0}$, which further gives $\mathbf{W}_0 = \mathbf{0}$ and $\text{Tr}(\mathbf{W}_0) = 0$. Accordingly, the general transmit-power constraint in (2) reduces to $\sum_{k=1}^K \|\mathbf{w}_k\|_2^2 \leq P_{\max}$.

以及：

However, in this work, the target location is unknown. Therefore, the target-related geometric parameters, such as the RIS–target distance and angle, cannot be directly obtained. As a result, the RIS–target channel $\mathbf{h}$ and the corresponding round-trip sensing channel $\mathbf{B}(\boldsymbol{\theta})$ are unknown, which makes the sensing-EE maximization problem more challenging.

这一版不是简单说“审稿人说错了”，而是说明：